

Rethinking the Geometry of Power: A Review of Vertical GaN Devices, Materials, and Physics

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As we approach the theoretical limits of Silicon, the power electronics community faces a geometric dilemma. While Lateral GaN HEMTs have successfully commercialized the low-voltage domain, their surface-centric architecture struggles with the high-voltage (> 1.2 kV) and high-power demands of next-generation AI data centers and EV powertrains. This paper adopts a reflective approach to Vertical GaN (v-GaN) technology. We explore the fundamental physics that necessitate a shift from lateral to vertical transport—specifically the decoupling of breakdown voltage from chip area. We critically examine the material bottlenecks (native vs. foreign substrates), the "Achilles' heel" of p-type doping, and the latest architectural innovations (CAVET, FinFET). Finally, we assess the 2024-2025 breakthroughs in hybrid integration and edge termination that are moving v-GaN from laboratory curiosity to industrial inevitability.

Index Terms—Vertical GaN, Physics of Failure, Epitaxial Growth, Power MOSFET, Edge Termination, FIT.

I. INTRODUCTION: THE GEOMETRIC IMPERATIVE

WHEN one considers the trajectory of power electronics, it becomes evident that we are fighting a battle against material limits. Silicon has served us well, but its narrow bandgap (1.1 eV) is a fundamental constraint. While Silicon Carbide (SiC) has bridged the gap to high voltage, Gallium Nitride (GaN) offers an even more tantalizing prospect: a critical electric field ($E_c \approx 3.3$ MV/cm) that theoretically allows for devices with superior Baliga Figures of Merit (BFOM).

However, the current dominance of Lateral GaN HEMTs presents a paradox. By confining current to a 2DEG at the surface, we inherently limit the device's ability to handle high electric fields and dissipate heat.

The question, then, is not just about material, but about geometry. Why force electrons to travel laterally

across a surface prone to traps and breakdown, when the bulk material offers a pristine vertical path?

Moving to a vertical architecture (v-GaN) is not merely a design choice; it is a physical imperative for high-power scaling. In v-GaN, the breakdown voltage (BV) scales with drift layer thickness (d), not surface area. This decoupling allows us to push voltages beyond 1200 V without expanding the die size, while simultaneously burying the peak electric field within the bulk to enable true avalanche capability—a feat impossible for lateral counterparts.

II. THE SUBSTRATE CONUNDRUM

If the physics of v-GaN are so superior, why has it not yet dominated the market? The answer lies in the crystal lattice itself.

A. The Ideal vs. The Practical

In an ideal world, we would rely solely on Homoepitaxy (GaN-on-GaN).

Growing GaN on its native substrate eliminates lattice mismatch and thermal expansion issues, yielding dislocation densities as low as 10^3 cm^{-2} . This is critical, as leakage mechanisms in vertical devices are often dominated by Variable-Range Hopping (VRH) through these defects.

However, we must confront the economic reality: native GaN substrates remain prohibitively expensive.

B. The Heteroepitaxial Compromise

This forces us to explore Heteroepitaxy. We attempt to grow vertical structures on foreign substrates like Silicon or Sapphire, but this is a fight against nature.

- **GaN-on-Si:** Economically attractive, yet the ~17% lattice mismatch creates a tension that often results in wafer bowing or cracking during the thick epitaxial growth required for high voltage.
- **GaN-on-Sapphire:** While cheaper, its poor thermal conductivity ($0.25 \text{ W/cm}\cdot\text{K}$) acts as a thermal bottleneck, negating the efficiency gains of GaN.

The industry is thus at a crossroads: optimize the cost of native substrates, or engineer the stress out of foreign ones?

III. PROCESSING: THE ART OF CONTROL

Constructing a vertical device requires mastering two notoriously difficult processes in GaN: activating holes and carving the crystal without killing it.

A. The P-Type Doping Challenge

P-type doping is arguably the "Achilles' heel" of GaN technology. Magnesium (Mg) is our primary acceptor, but its high activation energy means that at room temperature, most dopants remain inert.

We are essentially fighting thermodynamics. High Mg concentrations often lead to self-compensation defects. However, the 2024 breakthrough in **Mg-intercalated Superlattices (MiGs)** offers a new perspective. By engineering monolayer atomic sheets of Mg, we can potentially decouple mobility from concentration, creating a "super-highway" for holes that could revolutionize contact resistance.

B. Etching with Precision

To create structures like Trench MOSFETs, we must etch deep into the drift region. Standard ICP etching is too aggressive; it leaves behind a chaotic surface of nitrogen vacancies and plasma damage. The solution seems to lie in a hybrid approach: using dry etching for the shape, followed by wet TMAH etching to "heal" the sidewalls, aligning them to stable crystallographic planes to maximize channel mobility.

IV. ARCHITECTURAL EVOLUTION

How do we best utilize this vertical path? Three topologies stand out, each representing a different trade-off.

A. CAVET: The Hybrid Approach

The Current Aperture Vertical Electron Transistor (CAVET) is a fascinating hybrid. It attempts to marry the high mobility of the AlGaN/GaN heterojunction (from HEMTs) with a vertical drift region.

It avoids the p-doping channel issue but introduces a new challenge: fabricating a reliable Current Blocking Layer (CBL) buried within the device. It is a complex dance of regrowth and interface management.

B. Trench MOSFET: The Workhorse

The Trench MOSFET is the structure most likely to succeed commercially because it mimics the proven Silicon superjunction. The key metric here is specific on-resistance ($R_{on,sp}$). If we can manage the gate dielectric reliability—preventing premature breakdown at the trench corners—this topology offers the best balance of performance and manufacturability.

C. FinFET: Ultimate Control

When we need absolute electrostatic control, we turn to the FinFET. By wrapping the gate around a narrow GaN fin, we suppress leakage and short-channel effects. It is geometrically complex, but physically superior.

V. MANAGING THE EDGE: THE FIELD FRONTIER

A vertical power device is, paradoxically, often limited not by its bulk capabilities, but by its boundaries. The geometric transition from the active area

to the device periphery creates a discontinuity in the electric field. Without intervention, this "curvature effect" causes the field to crowd at the edge, leading to premature breakdown long before the drift layer reaches its theoretical avalanche limit. We are thus tasked with a geometric smoothing problem: how to spread the equipotential lines gracefully?

A. The Precision of JTE

The Junction Termination Extension (JTE) represents the classical solution. By implanting precise zones of p-type charge (P_1, P_2, P_3) with decreasing concentration outward from the main junction, we attempt to "flatten" the electric field peak.

However, in GaN, this is a precarious balancing act. The "JTE window"—the range of doping dosage that provides optimal shielding—is notoriously narrow. A slight deviation in implantation dose or activation efficiency can render the termination ineffective. We are essentially trying to paint a gradient with a discrete set of ion beams, where any error drastically reduces the blocking voltage.

B. The FIT Paradigm Shift (2025)

Here lies the elegance of the recent ****Fluorine-Implanted Termination (FIT)**** breakthrough. Rather than fighting the difficult battle of p-type activation (the Mg problem discussed earlier), researchers turned to the unique properties of Fluorine in GaN.

Fluorine ions, when implanted, are immobile and act as fixed negative charges. They effectively modulate the surface potential and expand the depletion region without requiring high-temperature annealing or complex p-GaN regrowth. This technique does not just improve performance; it simplifies the process integration, offering a high-yield path to breakdown voltages exceeding 1.2 kV. It transforms a complex doping problem into a simpler electrostatic engineering task.

VI. RECENT BREAKTHROUGHS (2024-2025)

The years 2024 and 2025 have marked a transition from exploring "what is possible" to demonstrating

"what is manufacturable." The narrative has shifted from record-breaking numbers to structural innovation.

A. The Hybrid Integration: GaN-on-SiC

The industry has long debated: GaN for speed, SiC for voltage/thermals. The recent emergence of hybrid GaN-on-SiC devices asks: "Why choose?" By bonding a high-quality GaN channel layer onto a SiC drift/substrate stack, we create a device that utilizes SiC's superior thermal conductivity ($3\times$ that of Si, $1.5\times$ that of GaN) for bulk heat extraction, while leveraging GaN's high electron mobility for the switching channel. This architecture effectively bypasses the thermal bottleneck of GaN-on-Si and the mobility limits of SiC MOSFETs. It is a pragmatic synthesis, acknowledging that no single material is perfect.

B. Shattering the Cost Barrier: Vertical GaN-on-Si

Perhaps the most disruptive development is the successful fabrication of fully vertical GaN MOSFETs on 8-inch Silicon wafers by the Ziener team. Historically, the lattice mismatch (-17%) made thick drift layers on Silicon crack. The breakthrough lies in stress-balancing buffer layers that allow for crack-free growth of thick ($> 5\mu m$) drift regions. This places v-GaN on a cost trajectory similar to standard Silicon power devices, potentially democratizing high-performance power electronics for cost-sensitive markets.

C. Optically Triggered FinFETs: Eliminating the Gate Loop

MIT's demonstration of optically switched vertical FinFETs addresses a fundamental system-level issue: Electromagnetic Interference (EMI).

In high-frequency switching (> 1 MHz), the electrical gate loop acts as an antenna for noise, leading to false triggering. By replacing the electrical gate driver with an optical signal, we achieve perfect galvanic isolation. The device is controlled by photons, not electrons, decoupling the control plane from the noisy power plane. This is not just a device innovation; it is a circuit topology revolution.

VII. APPLICATIONS: THE SYSTEM LEVEL IMPACT

Why do we endure these material and processing challenges? Because the system-level benefits are transformative. The value of v-GaN is not in the device itself, but in what it allows the system to become.

A. AI Data Centers: The Density Imperative

Modern AI processors are approaching power demands that threaten to overwhelm data center infrastructure. The challenge is converting 48V (or higher) bus voltages down to $< 1V$ at the point of load (PoL) with massive current capability. Vertical GaN, with its vertical current flow, enables **Integrated Voltage Regulators (IVR)** stacked directly on top of processors. Its high switching frequency shrinks passive components, freeing up precious board real estate for compute cores. Here, v-GaN is not just a switch; it is an enabler of Moore's Law.

B. Automotive Powertrains: Reliability First

For the 800V Electric Vehicle (EV) architecture, efficiency is mandatory, but reliability is non-negotiable. This is where v-GaN distinguishes itself from lateral GaN. In the event of a voltage spike from the motor, a lateral device typically fails catastrophically. A vertical device, however, can enter avalanche breakdown, dissipating the surge energy within its bulk volume. This "ruggedness" allows automotive designers to reduce safety margins and protective components, lightening the On-Board Charger (OBC) and extending vehicle range.

VIII. CONCLUSION

The journey of Vertical GaN is a testament to the persistence of physics over geometry. While lateral devices provided the initial spark for the GaN revolution, the inherent geometric advantages of vertical transport—area-independent breakdown, bulk thermal paths, and true avalanche capability—make v-GaN the inevitable successor for the high-power domain.

We are currently witnessing the convergence of critical technologies: **MiGs** have unlocked p-type conductivity; **FIT** has tamed the edge fields;

and **Heteroepitaxy** is solving the cost equation. The transition from "lab curiosity" to "industrial workhorse" is underway. As we look toward a future electrification powered by AI and renewable energy, Vertical GaN stands not merely as an alternative to Silicon or SiC, but as the essential architecture required to bridge the gap between high efficiency and high power density.

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